Akihiro Miki

Project Assistant Professor

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Summary

- a Project Assistant Professor at the University of Tokyo, working on the intersection of biological understanding and intelligent machine design, with a focus on biomimetic robotics to investigate biological systems and their underlying principles.

Research Interests

Biomimetic Robotics, Soft Robotics, Musculoskeletal Humanoid, biological System, Neuroscience, Physiology, Embodied Intelligence, Constructive Understanding, Complex Emergent Systems

Experience

The University of Tokyo, Project Assistant Professor — Tokyo, Japan
Working at JSK Robotics Laboratory (Creative Informatics, Information Science and Technology). — Apr 2026 – present, 1 month

JSPS (Japan Society for the Promotion of Science), JSPS Research Fellowships for young students (DC2) — Tokyo, Japan
Apr 2024 – Mar 2026, 2 years 1 month

Japan Science and Technology Agency (JST), The University of Tokyo "Advanced Human Resource Development Leading Green Transformation (GX) (SPRING GX)" Project student — Tokyo, Japan
Apr 2023 – Mar 2024, 1 year 1 month
- Description 1. JST Support for Pioneering Research Initiated by the Next Generation (SPRING) Program

The University of Tokyo, Research Assistant — Tokyo, Japan
May 2022 – Mar 2026, 4 years

Education

PhD — **The University of Tokyo**, Mechano-Informatics — Tokyo, Japan, Apr 2023 – Mar 2026
- PhD in Graduate School of Information Science and Technology
- Dissertation: A Cross-Hierarchical Constructive Framework for Biological Understanding Based on Reproducing Emergent Functional Processes via Biomimetic Tissue Robotics

Master — **The University of Tokyo**, Interdisciplinary Information Studies — Tokyo, Japan, Apr 2021 – Mar 2023
- Completed Master's program at Graduate School of Interdisciplinary Information Studies

Bachelor — **The University of Tokyo**, Mechano-Informatics — Tokyo, Japan, Apr 2017 – Mar 2021
- Graduated from Department of Mechano-Informatics, Faculty of Engineering

M.D. Program (withdrawn)	**Ehime University**, Medicine	Ehime, Japan Mar 2017

Awards

- 2026-03-24 - Valedictorian (Graduation Ceremony Representative), The University of Tokyo (link)
- 2025-03-18 - 2024 SICE Young Authors Award for Basic Research, The Society of Instrument and Control Engineers (link)
- 2024-01-23 - The 24th SICE Systems Integration Conference Excellent Presentation Award, The Society of Instrument and Control Engineers (link)
- 2023-09-13 - The 3th Robotics Symposia Excellent Research and Technology Award, The Robotics Society of Japan (link)

Grants-in-Aid and Scholarship

- 2024-04 to 2026-03 - JSPS Research Fellowship for Young Scientists (DC2), JSPS (Japan Society for the Promotion of Science)
- 2023-04 to 2024-03 - The University of Tokyo "Advanced Human Resource Development Leading Green Transformation (GX) (SPRING GX)" Project student., Japan Science and Technology Agency (JST)
- 2023-04 to 2024-03 - Iwai Hisao Memorial Tokyo Scholarship Foundation
- 2022-04 to 2023-03 - HAYASHI Rheology Scholarship Foundation
- 2021-04 to 2022-03 - Iwadare Scholarship Foundation Scholarship
- 2019-04 to 2021-03 - HAYASHI Rheology Scholarship Foundation
- 2019-04 to 2023-03 - SINNIHON SCHOLARSHIP FOUNDATION Scholarship

Media

- 2026-03-24 - AY 2025 Diploma Presentation Ceremony held (Valedictorian), The University of Tokyo (link)

International Journal Papers (4)

Exploring the proprioceptive potential of joint receptors using a biomimetic robotic joint — Jan 2026
Akihiro Miki, Shun Hasegawa, Sota Yuzaki, Yuta Sahara, Yoshimoto Ribayashi, Kento Kawaharazuka, Kei Okada
https://doi.org/10.1038/s41598-025-27311-3 (Scientific Reports, vol. 16, no. 1, pp. 4724)

Liquid Metal Sloshing for High-Load Active Self-Healing System: An Application to Tendon-Driven Legged Robot — Jan 2025
Shinsuke Nakashima, Kento Kawaharazuka, Yuya Nagamatsu, Koki Shinjo, Akihiro Miki, Yuki Asano, Yohei Kakiuchi, Kei Okada, Masayuki Inaba
https://doi.org/10.1002/aisy.202500040 (Advanced Intelligent Systems, vol. 7, no. 7, pp. 2500040)

Adaptive Body Schema Learning System Considering Additional Muscles for Musculoskeletal Humanoids — Jan 2022
Kento Kawaharazuka, Akihiro Miki, Yasunori Toshimitsu, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/LRA.2022.3147457 (IEEE Robotics and Automation Letters, vol. 7, no. 2, pp. 3459-3466)

Dynamic Cloth Manipulation Considering Variable Stiffness and Material Change Using Deep Predictive Model With Parametric Bias — Jan 2022
Kento Kawaharazuka, Akihiro Miki, Masahiro Bando, Kei Okada, Masayuki Inaba
https://doi.org/10.3389/fnbot.2022.890695 (Frontiers in Neurorobotics, vol. 16, pp. 1-16)

International Conference Proceedings (Peer Reviewed) (16)

Construction of Musculoskeletal Simulation for Shoulder Complex with Ligaments and Its Validation via Model Predictive Control — Jan 2024
Yuta Sahara, Akihiro Miki, Yoshimoto Ribayashi, Shunnosuke Yoshimura, Kento Kawaharazuka, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS58592.2024.10802465 (Proceedings of The 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 327-333)

Designing Fluid-Exuding Cartilage for Biomimetic Robots Mimicking Human Joint Lubrication Function — Jan 2024
Akihiro Miki, Yuta Sahara, Kazuhiro Miyama, Shunnosuke Yoshimura, Yoshimoto Ribayashi, Kento Kawaharazuka, Shun Hasegawa, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/RoboSoft60065.2024.10521920 (Proceedings of 2024 IEEE-RAS International Conference on Soft Robotics, pp. 452-459)

Fundamental Three-Dimensional Configuration of Wire-Wound Muscle-Tendon Complex Drive — Jan 2024
Yoshimoto Ribayashi, Yuta Sahara, Shogo Sawaguchi, Kazuhiro Miyama, Akihiro Miki, Kento Kawaharazuka, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/Humanoids58906.2024.10769901 (Proceedings of the 2024 IEEE-RAS International Conference on Humanoid Robots, pp. 980-987)

Patterned Structure Muscle : Arbitrary Shaped Wire-driven Artificial Muscle Utilizing Anisotropic Flexible Structurefor Musculoskeletal Robots — Jan 2024
Shunnosuke Yoshimura, Akihiro Miki, Kazuhiro Miyama, Yuta Sahara, Kento Kawaharazuka, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS58592.2024.10801899 (Proceedings of The 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 13930-13937)

Vlimb: A Wire-Driven Wearable Robot for Bodily Extension, Balancing Powerfulness and Reachability — Jan 2024
Shogo Sawaguchi, Temma Suzuki, Akihiro Miki, Kento Kawaharazuka, Sota Yuzaki, Shunnosuke Yoshimura, Yoshimoto Ribayashi, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/Humanoids58906.2024.10769893 (Proceedings of the 2024 IEEE-RAS International Conference on Humanoid Robots, pp. 851-857)

Development of a Wire-Wound Muscle-Tendon Complex Drive and Its Application to a Two-Dimensional Robot Configuration — Jan 2023
Yoshimoto Ribayashi, Kazuhiro Miyama, Akihiro Miki, Kento Kawaharazuka, Kei Okada, Koji Kawasaki, Masayuki Inaba
https://doi.org/10.1109/Humanoids57100.2023.10375220 (Proceedings of the 2023 IEEE-RAS International Conference on Humanoid Robots, pp. 758-764)

Fault Mitigation and Fault Tolerant Humanoid System against Contact Impact — Jan 2023
Takuma Hiraoka, Shimpei Sato, Naoki Hiraoka, Akihiro Miki, Kunio Kojima, Kei Okada, Masayuki Inaba, Koji Kawasaki
https://doi.org/10.1007/978-3-031-44851-5_27 (Intelligent Autonomous Systems 18, pp. 353-366)

Fusion of Body and Environment with Movable Carabiners for Wire-Driven Robots Toward Expansion of Physical Capabilities — Jan 2023
Sota Yuzaki, Akihiro Miki, Masahiro Bando, Shunnosuke Yoshimura, Temma Suzuki, Kento Kawaharazuka, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/Humanoids57100.2023.10375200 (Proceedings of the 2023 IEEE-RAS International Conference on Humanoid Robots, pp. 679-685)

Muscle-Tendon Complex-Inspired Deformable Exteriors as a Wire-Drive Extension Jan 2023
Yoshimoto Ribayashi, Kazuhiro Miyama, Akihiro Miki, Kento Kawaharazuka, Kei Okada, Koji Kawasaki, Masayuki Inaba
https://doi.org/10.18910/92308 (Proceedings of 11th International Symposium on Adaptive Motion of Animals and Machines, pp. P53)

System Architecture and Real-World Task Realization of Musculoskeletal Wheeled Robot Musashi-W with Various Hardware Components Jan 2023
Akihiro Miki, Kento Kawaharazuka, Masahiro Bando, Kei Okada, Koji Kawasaki, Masayuki Inaba
https://doi.org/10.1007/978-3-031-44851-5_9 (Intelligent Autonomous Systems 18, pp. 109-122)

DIJE: Dense Image Jacobian Estimation for Robust Robotic Self-Recognition and Visual Servoing Jan 2022
Yasunori Toshimitsu, Kento Kawaharazuka, Akihiro Miki, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS47612.2022.9981868 (Proceedings of The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 2219-2226)

Design of Robot Foot with Outer Edge Measurement Structure and Chair Rotation Motion by Friction Control Jan 2022
Yoshimoto Ribayashi, Kento Kawaharazuka, Yasunori Toshimitsu, Daiki Kusuyama, Akihiro Miki, Koki Shinjo, Masahiro Bando, Temma Suzuki, Yuta Kojio, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/Humanoids53995.2022.10000127 (Proceedings of the 2022 IEEE-RAS International Conference on Humanoid Robots, pp. 314-321)

Hardware Design and Learning-Based Software Architecture of Musculoskeletal Wheeled Robot Musashi-W for Real-World Applications Jan 2022
Kento Kawaharazuka, Akihiro Miki, Masahiro Bando, Temma Suzuki, Yoshimoto Ribayashi, Yasunori Toshimitsu, Yuya Nagamatsu, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/Humanoids53995.2022.10000123 (Proceedings of the 2022 IEEE-RAS International Conference on Humanoid Robots, pp. 413-419)

Imitation Behavior of the Outer Edge of the Foot by Humanoids Using a Simplified Contact State Representation Jan 2022
Yoshimoto Ribayashi, Kento Kawaharazuka, Yasunori Toshimitsu, Daiki Kusuyama, Akihiro Miki, Koki Shinjo, Masahiro Bando, Temma Suzuki, Yuta Kojio, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS47612.2022.9981673 (Proceedings of The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 4243-4249)

Learning of Balance Controller Considering Changes in Body State for Musculoskeletal Humanoids Jan 2022
Kento Kawaharazuka, Yoshimoto Ribayashi, Akihiro Miki, Yasunori Toshimitsu, Temma Suzuki, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS47612.2022.9981051 (Proceedings of The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 5809-5816)

RAMIEL: A Parallel-Wire Driven Monopedal Robot for High and Continuous Jumping Jan 2022
Temma Suzuki, Yasunori Toshimitsu, Yuya Nagamatsu, Kento Kawaharazuka, Akihiro Miki, Yoshimoto Ribayashi, Masahiro Bando, Kunio Kojima, Yohei Kakiuchi, Kei Okada, Masayuki Inaba
https://doi.org/10.1109/IROS47612.2022.9981963 (Proceedings of The 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 5017-5024)